Natural Deduction Rules (System F)

System F (from *Language, Proof, and Logic*) consists of the following inference rules. Each rule specifies how a conclusion may be derived from given premises. Expressions of the form $P \vdash Q$ indicate a **subproof**: assume P, derive Q, and then discharge the assumption.

And Introduction

($\wedge$ Intro)

$$\frac{P_1,\ \ldots,\ P_n}{\vdash\ P_1 \wedge \cdots \wedge P_n}$$

And Elimination

($\wedge$ Elim)

$$\frac{P_1 \wedge \cdots \wedge P_n}{\vdash\ P_i}$$

Or Introduction

($\vee$ Intro)

$$\frac{P_i}{\vdash\ P_1 \vee \cdots \vee P_n}$$

Or Elimination

($\vee$ Elim)

$$\frac{P_1 \vee \cdots \vee P_n,\ P_1 \vdash Q,\ \ldots,\ P_n \vdash Q}{\vdash\ Q}$$

Negation Introduction

($\neg$ Intro)

$$\frac{P \vdash Q,\ P \vdash \neg Q}{\vdash\ \neg P}$$

Negation Elimination

($\neg$ Elim)

$$\frac{\neg\neg P}{\vdash\ P}$$

Implication Introduction

($\rightarrow$ Intro)

$$\frac{P \vdash Q}{\vdash\ P \rightarrow Q}$$

Implication Elimination

($\rightarrow$ Elim)

$$\frac{P \rightarrow Q,\ P}{\vdash\ Q}$$

Biconditional Introduction

($\leftrightarrow$ Intro)

$$\frac{P \vdash Q,\ Q \vdash P}{\vdash\ P \leftrightarrow Q}$$

Biconditional Elimination

($\leftrightarrow$ Elim)

$$\frac{P \leftrightarrow Q}{\vdash\ P \rightarrow Q,\ Q \rightarrow P}$$

Reiteration: From P, infer P.

Notes: $\vee$ Elim corresponds to *proof by cases*. $\neg$ Intro corresponds to *proof by contradiction* (reductio). $\rightarrow$ Intro corresponds to *conditional proof*.